

Luca Tomei

Software Engineer · IT Development Advisor

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Professional Summary

Software Engineer (MSc Engineering in Computer Science, Sapienza University of Rome) building enterprise systems since 2021: full-stack development with .NET 8, Angular and SQL Server, Infor M3 ERP integration (on-premise and cloud) through ION, MEC and ESB, microservices, REST APIs and low-code delivery on OutSystems.

IT Development Advisor at Yokohama TWS, hands-on in a 4-person IT team: development standards, CI/CD and containerisation strategy, about 15 integration flows with 3PL partners, technical oversight of outsourced work. End-to-end ownership from requirements to deployment and support. Cybersecurity enthusiast: firmware reverse engineering, NFC/RFID research on Flipper Zero, hardened self-hosted infrastructure.

Technical Skills

Languages: C#, Python, TypeScript, JavaScript, Java, SQL, PHP, C/C++, Scala, Swift, Kotlin, PowerShell

Backend & APIs: .NET 8, Laravel, Node.js, REST APIs, Microservices, RabbitMQ, XSD/XML, JSON

Frontend & Mobile: Angular, Vue.js, React, React Native, SwiftUI

ERP & Integration: Infor M3 (on-premise and cloud), Infor ION, MEC, ESB, 3PL integrations, OutSystems

Databases & Data: SQL Server, MySQL, PostgreSQL, MongoDB, SQLite, Databricks

DevOps & Cloud: Azure DevOps, GitHub Actions, CI/CD, Docker, Linux, IIS, Proxmox, Cloudflare, Azure

Embedded & Security: Firmware reverse engineering, NFC/RFID (MIFARE Classic), Flipper Zero, ESP32, MQTT, Home Assistant, Node-RED, network and VPN configuration

Design & CAD: Autodesk Fusion 360, AutoCAD, 3D printing

Ways of working: Agile, code review, requirements analysis, technical documentation, cross-functional collaboration, problem solving, mentoring

Work Experience

IT Development Advisor

February 2026 – Present

Yokohama TWS (formerly Trelleborg Wheel Systems) – Tivoli (Rome), Italy

- Define development strategy and best practices: microservices, Azure DevOps CI/CD pipelines, Docker-based deployments
- Develop hands-on within a 4-person IT team (one manager, three advisors), owning code quality and performance on .NET 8, Angular and SQL Server
- Design and govern enterprise API orchestration between microservices and legacy systems via Infor ION, MEC and ESB
- Coordinate the Infor M3 cloud migration and the rollout of OutSystems low-code applications, two of them already in production
- Own about 15 integration flows with 3PL logistics partners and the technical oversight of outsourced developments; turn business requirements into documented technical specifications

IT Development Specialist

May 2023 – February 2026

- Led technical analysis, development and implementation of custom solutions on Infor M3 ERP (on-premise) and OutSystems
- Developed REST APIs and C#/.NET backend microservices integrated with Angular dashboards and SQL Server databases
- Designed and maintained about 15 enterprise integration flows (MEC, ESB) with 3PL partners: XSD/XML schemas, data mapping, troubleshooting of inbound/outbound documents
- Built Azure DevOps CI/CD pipelines replacing manual multi-step releases with automated, repeatable deployments of frontend, backend and database
- Performed code reviews and root cause analysis; maintained technical documentation and runbooks

Stack: .NET 8, C#, Angular, TypeScript, SQL Server, MySQL, PHP/Laravel, Vue.js, RabbitMQ, Infor M3, Infor ION, MEC, ESB, OutSystems, Azure DevOps, Docker, IIS

ERP Developer

November 2021 – May 2023

Trelleborg Wheel Systems – Tivoli (Rome), Italy

- Managed the technical side of the Infor M3 ERP (Business Engine, interfaces, layouts, reports) and its integration with other business systems
- Analysed functional documentation, wrote technical documentation, fixed bugs and kept ERP versions aligned with vendor releases
- Supported the business on technical ERP issues and worked with vendor consultants on complex implementations

Teaching Assistant, Functional and Parallel Programming

October 2021 – December 2021

Sapienza University of Rome – Rome, Italy

- Tutored the online section (about 100 students) of the Scala-based course held by Prof. Camil Demetrescu: labs, programming assignments, quizzes and exams

Software Developer Intern – OBDA Systems, Rome (web front end: React, REST)

July 2018 – September 2018

Home Automation and IT Technician (freelance)

November 2015 – October 2021

Self-employed – Rome, Italy

- Installed home automation and control systems for homes and shops; built databases for business data management; provided technical consulting

Education

Master's Degree in Engineering in Computer Science – 110/110

January 2020 – October 2021

Sapienza University of Rome – Rome, Italy

- Thesis: *Design and proof-of-concept of the integration of an electric scooter with a wheelchair: from the mechanical design to the firmware upgrade* – custom firmware with adaptive acceleration, Python client/GUI for firmware builds, 3D-designed bracket

Bachelor's Degree in Computer and Automation Engineering – 99/110

2016 – 2019

Sapienza University of Rome – Rome, Italy

- Thesis: *Mapping Management of an Ontology-based Data Access System* (Java, React)

Scientific High School Diploma – Liceo Scientifico Ettore Majorana, Guidonia (Rome)

2009 – 2014

Projects

Electric Scooter Wheelchair – github.com/LucaTomei/Electric-Scooter-Wheelchair

2020 – 2021

- Built a working prototype coupling an electric scooter to a wheelchair: 3D-printed CAD bracket and reverse-engineered firmware with acceleration and braking tuned for the new load (MSc thesis) (Python, firmware, Fusion 360, 3D printing)

OPBG Telegram Booking Bot

Private project

- Built an invite-only multi-user bot automating the Bambino Gesù children's hospital booking portal: availability monitoring with notification or auto-booking, separate bot and worker processes, idempotent booking protocol, admin panel (Python, aiogram, SQLAlchemy, MySQL, Docker)

Flipper Mi Band NFC Writer – github.com/LucaTomei/flipper-miband-nfc-writer

2025

- Developed a Flipper Zero app that reads, writes, emulates and verifies NFC data on Xiaomi Mi Band devices: automatic backups, sector-level progress, hex diff viewer, persistent settings and exportable logs (C, Flipper Zero, NFC, MIFARE Classic)

Telegram Audio Transcriber – github.com/LucaTomei/Telegram-Audio-Transcriber

2026

- Built a self-hosted bot transcribing voice messages and audio locally with Whisper models; optional text cleanup via OpenAI-compatible endpoints, whitelist and admin roles, no audio persistence, Docker Compose deployment from CI-built images (Python, Faster-Whisper, ffmpeg, SQLAlchemy, Docker, GitHub Actions)

LazioHealthMonitorBot – github.com/LucaTomei/LazioHealthMonitorBot

2025 – Present

- Created a bot used by about 35 people to poll the Lazio regional health service every five minutes: real-time notifications, filters, direct or automatic booking of the first free slot, PDF confirmations, CI and container images on every release (Python, Telegram Bot API, REST, Docker, GitHub Actions)

Carburanti Bot – github.com/LucaTomei/Carburanti-Bot

2026

- Developed a real-time fuel price bot for Italy with about 12 active users: favourite station, nearby and best-price search from a shared location, daily notifications, keyboard UI and admin panel (Python, Docker, Portainer, MIMIT/Osservaprezzi open data)

Estrai Allegati – github.com/LucaTomei/estrai-allegati

2026

- Developed a native macOS app extracting embedded files from Office documents: dependency-free ZIP, OLE2/Compound File and EMF parsers written from scratch, universal binary, automated DMG releases (Swift, SwiftUI, GitHub Actions)

Computer Scientists Retrieval – github.com/LucaTomei/Computer_Scientists

2020

- Ranked computer scientists by influence with PageRank and HITS over a Wikipedia/DBpedia graph, with category classification (Python, NetworkX, SPARQL/DBpedia, BeautifulSoup)

Other open source

2019 – Present

- **MAXG30CFW** – Ninebot Max G30 firmware builder (Python, reverse engineering) · **telegram-ha-monitor** – multi-server monitoring bot with leader election and RBAC · **Esp32-Cam-PIR-Telegram** – motion-triggered camera controlled via Telegram (C++) · **Hoodie** (iOS, Swift) · **ColliGO** – queue-management Telegram bot for shops during COVID-19 (freelance, 2020)

Certifications

OutSystems Associate Reactive Developer – OutSystems, 2024

Academy Accreditation – Databricks Fundamentals (2025); Databricks Lakehouse Fundamentals (2022) – Databricks

Software Architecture Fundamentals; Artificial Intelligence Fundamentals: Machine Learning; Cloud Computing: Core Concepts – Birdify – LMSAFP, 2024

Security: Introduction to Cybersecurity Tools & Cyber Attacks (IBM, Coursera, 2022); Cybersecurity Foundations, Cybersecurity Threat Landscape, Cloud Security Awareness, DevOps Foundations: DevSecOps (LinkedIn Learning, 2024)

Introducing Semantic Kernel: Building AI-Based Apps; Agile Software Development; APIs and Web Services; Test Automation;

Behavior-Driven Development; Project Management Foundations; Data Engineering; Generative AI and Prompt Engineering – LinkedIn Learning, 2024

Introduction to iOS Mobile Application Development – Meta, Coursera, 2023

Foundations of Project Management; Foundations of User Experience (UX) Design – Google, Coursera, 2022

Introduction to Cloud Computing; Introduction to Data Analytics – IBM, Coursera, 2022

Crash Course on Python; Using Python to Interact with the Operating System – Google, Coursera

REST API (Intermediate); SQL (Basic, Intermediate); Java, Python, React (Basic) – HackerRank, 2021

Languages

Italian – Native · English – B2 (CEFR) · French – A2 (CEFR)

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